

(*) Let A be any set.

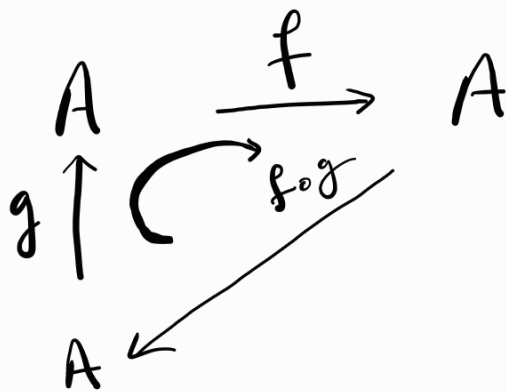
$$G = \{ f: A \rightarrow A \mid f \text{ is a bijection } \}$$

Prove that $(G, \circ)$ is a group.

pf: ① Closure: let f, g $\in G$

we need, f $\circ$ g $\in G$

as f, g are bijective fun,



f $\circ$ g is also bijective.

We first show that f $\circ$ g is 1-1:

$$\underline{f \circ g}(x) = \underline{f \circ g}(y) \quad \text{for } x, y \in A$$

$$\Rightarrow \underline{f}(g(x)) = \underline{f}(g(y)) \quad , \quad \underline{f \text{ is bijective}} \Rightarrow f \text{ is 1-1}$$

$$\Rightarrow \underline{g(x)} = \underline{g(y)}$$

$$\Rightarrow \underline{x} = \underline{y} \Rightarrow \underline{f \circ g} \text{ is 1-1}$$

(*) f $\circ$ g is onto:

$f \circ g : A \rightarrow A$ f is bijective $\Rightarrow f$ is onto.

Let $y \in A$,
From ①

$$f(x) = y$$

$$\Rightarrow f(g(x)) = y \text{ [from ②]}$$

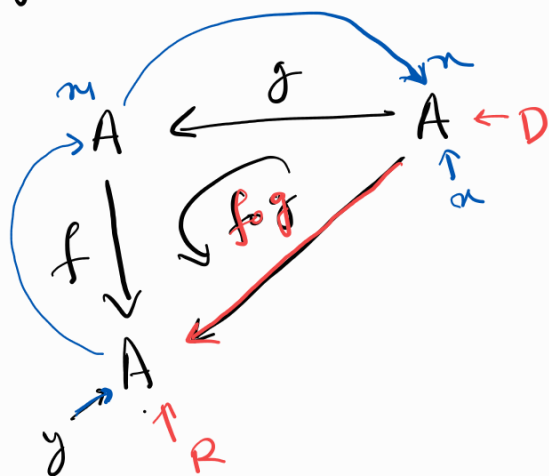
$$\Rightarrow f \circ g(x) = y.$$

$$\exists x_1 \in A \text{ s.t. } f(x_1) = y. \text{ --- ①}$$

$$f : A \rightarrow A \quad \begin{matrix} \uparrow x \\ \uparrow y \end{matrix}$$

$$g : A \rightarrow A \quad \begin{matrix} \uparrow x \\ \uparrow x_1 \end{matrix}$$

$$g(x) = x_1 \text{ --- ②}$$



$$(f \circ g)^{-1} = g^{-1} \circ f^{-1}$$

$$\Rightarrow f \circ g \in G$$

$$\text{② } f, g, h \in G, \quad f \circ (g \circ h) = (f \circ g) \circ h. \text{ (why?)}$$

$$\forall x \in A, \quad \underline{f \circ (g \circ h)}(x) = f((g \circ h)(x))$$

$$= f(g(h(x)))$$

$$\underline{(f \circ g) \circ h}(x) = (f \circ g)(h(x))$$

$$= f(g(h(x)))$$

$$\Rightarrow \underline{f \circ (g \circ h)} = \underline{(f \circ g) \circ h}$$

$$\text{③ } Id_A : A \rightarrow A$$

$$x \mapsto x$$

$$Id_A(x) = x, \quad Id_A \in G.$$

we want to show $f \circ \text{Id}_A = f$

$$\underline{f \circ \text{Id}_A}(\underline{x}) = f(\underline{\text{Id}_A(x)}) = f(\underline{x})$$

$$\Rightarrow \underline{f \circ \text{Id}_A = f}$$

$$(Id_A \circ f)(x) = \underline{Id_A(f(x))} = \underline{f(x)}$$

$$\Rightarrow \boxed{Id_A \circ f = f = f \circ Id_A}$$

(iv) f bijective map, $\exists$ $f^{-1} : A \rightarrow A$.

$$f \circ f^{-1}(x) = \underline{f(f^{-1}(x))} = x = \text{Id}_A(x)$$

$$\Rightarrow f \circ f^{-1} = \text{Id}_A$$

$$f^{-1} \circ f(a) = \underline{f^{-1}(f(a))} = a = \text{Id}_A(a)$$

$$\Rightarrow f^{-1} \circ f = \text{id}_A.$$

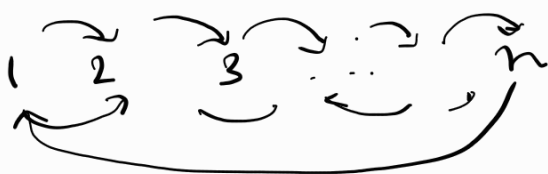
$(G_2 = \{ f: A \rightarrow A \mid f \text{ is bijection} \}, \circ)$ is a group.

$\downarrow$
any set

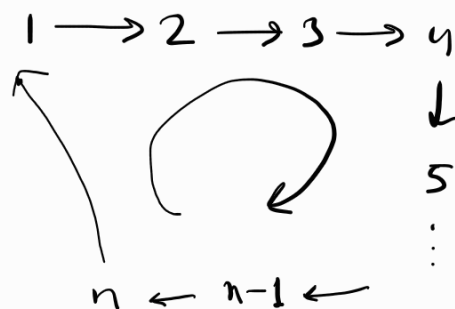
- particularly, $A = \{ \underbrace{1, 2, 3, \dots, n} \} = \{ a_1, a_2, \dots, a_n \}$

$$G_2 = \{ f: A \rightarrow A \mid f \text{ is bijection} \} = \underline{S_n} \text{ (Symmetric group / permutation group on } n \text{ element set)}$$

- $\text{order}(S_n) = |S_n| = o(S_n) \doteq n!$



$$\sigma \in S_n \quad \begin{aligned} \sigma(1) &= 2 \\ \sigma(2) &= 3 \\ &\vdots \\ \sigma(n) &= 1 \end{aligned}$$



$$\sigma = \begin{pmatrix} 1 & 2 & 3 & \dots & n-1 & n \\ 2 & 3 & 4 & \dots & n & 1 \end{pmatrix}$$

$$\rho = \begin{pmatrix} 1 & 2 & 3 & 4 & \dots & n-1 & n \\ 2 & 1 & 4 & 3 & \dots & n & n-1 \end{pmatrix} \rightarrow n \text{ even}$$

$$S_3, \quad \rho = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$

$$\rho_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} = \rho_1^{-1}$$

$$\rho_2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

$$\rho_2^{-1} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$\begin{aligned} \rho_2(1) &= 2 \Rightarrow \rho_2^{-1}(2) = 1 \\ \rho_2(2) &= 3 \Rightarrow \rho_2^{-1}(3) = 2 \end{aligned}$$

$$\downarrow P_2(3) = 1 \Rightarrow P_2^{-1}(1) = 3$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$P_2 = \begin{pmatrix} \begin{matrix} 1 & 2 & 3 \\ \uparrow & \uparrow & \uparrow \\ 2 & 3 & 1 \end{matrix} \end{pmatrix}$$

$$P_2^{-1} = \begin{pmatrix} 2 & 3 & 1 \\ 1 & 2 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$P_3 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \begin{matrix} \curvearrowright \\ \curvearrowright \end{matrix}$$

$$P_3^{-1} = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 2 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

$$P_4 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} \begin{matrix} \curvearrowright^2 \\ \curvearrowright \end{matrix} \quad P_4^2 =$$

$$P_4^{-1} = \begin{pmatrix} 1 & 3 & 2 \\ 1 & 2 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

$$P_6 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \begin{matrix} 1 \rightarrow 3 \\ 3 \rightarrow 2 \\ 2 \rightarrow 1 \end{matrix}$$

$$P_6^{-1} = \begin{pmatrix} 3 & 1 & 2 \\ 1 & 2 & 3 \end{pmatrix} \rightarrow P_6^3 = \text{Id}$$

$$= \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

Th^m: Let G be a group and $g \in G$. Then $o(g) = o(g^{-1})$.

pf: Let $o(g) = n$, $o(g^{-1}) = m$, claim: $n = m$.

$$(g^{-1})^n = (g^n)^{-1} \quad [\text{From previous claim}]$$

$$\begin{aligned} (g^{-1})^m &= e = (e)^{-1} = e & (g^{-1})^n &= (g^n)^{-1} = e^{-1} = e \\ \Rightarrow (g^m)^{-1} &= e \Rightarrow g^m = e & \Rightarrow m \mid n. & \\ \Rightarrow g^n &= e & \Rightarrow n \mid m. & \\ \Rightarrow m &\geq n. & \Rightarrow n &\geq m. \\ \Rightarrow m &= n. & & \end{aligned}$$

[Let $x \in G$, if $o(x) = n$, then $\forall k \in \mathbb{N}$ s.t. $x^k = e \Leftrightarrow n \mid k$]

pf: $n \mid k \Rightarrow k = nq$, for some $q \in \mathbb{N}$.

$$\underline{x^k} = x^{nq} = (x^n)^q = (e)^q = \underline{e}.$$

$$\Rightarrow \underline{x^k} = e.$$

$$\boxed{n|k} \Rightarrow x^k = e$$

conversely, $(x^k = e)$

$o(x) = n \Rightarrow$ n is the smallest natural no such that

$$\underline{x^n = e}, \Rightarrow \underline{k \geq n} \Rightarrow n|k \Leftrightarrow$$

$$\Rightarrow \boxed{k = nq + r} \quad \text{①}$$

Euclidean's Division Algorithm

$$\rightarrow \boxed{0 \leq r < n}$$

$$x^k = e$$

$$\Rightarrow x^{(nq+r)} = e \Rightarrow x^{nq} \cdot x^r = e$$

$$\Rightarrow (x^n)^q \cdot x^r = e$$

$$\Rightarrow e^q \cdot x^r = e$$

$$\Rightarrow \underline{x^r = e} \Rightarrow \underline{r=0}$$

$$\left(\begin{array}{l} x^1 \cdot x^{-1} = x^0 \\ = x^0 \\ = e \end{array} \right)$$

From eqⁿ ①,

$$k = nq + 0 \Rightarrow k = nq \Rightarrow \boxed{n|k}$$

$(\mathbb{R}, +, \cdot)$

$$a_2x^2 + a_1x + a_0 = 0 \quad +$$

$$x = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_2a_0}}{2a_2}$$

$$(F, +, \cdot) \quad (E, +, \cdot)$$

$$a_3x^3 + a_2x^2 + a_1x + a_0 = 0$$

$$x = \dots$$

$+$, $\times$
 $\swarrow$ $\searrow$

$$\checkmark \quad a_n x^n + \dots + a_0 = 0$$

$$\rightarrow \frac{a_n x^n + \dots + a_0 = 0}{n \geq 5}$$

Group theory, Ring theory, Field
theory